

Rainfall Data Analysis Report

Project Objective

The objective of this project is to perform exploratory data analysis (EDA) on rainfall data using Python libraries such as Pandas, Matplotlib, and Seaborn. The project aims to identify rainfall patterns, handle missing values, detect and treat outliers, reduce skewness, and generate meaningful visual insights from the dataset.

Problem Statement

Rainfall datasets often contain:

- missing values
- highly skewed distributions
- extreme outliers
- seasonal variations

These issues affect statistical analysis and visualization accuracy. The purpose of this project is to clean and preprocess the rainfall dataset, improve data distribution, and analyze rainfall behavior using various visualization techniques.

Dataset Description

- Dataset Name: Rainfall Data Analysis Dataset
- Total Records: 204876 rows
- Dataset Type: Structured CSV dataset
- Source: Github

Dataset Column

Column	Description
Id	Unique identifier
Date	Date of rainfall observation
State Code	State code
Actual	Actual rainfall received
rfs	Rainfall forecast statistics
Normal	Normal rainfall values
Deviation	Difference between actual and normal rainfall
Month	Month extracted from date
Rainfall_category	Categorized rainfall intensity

Data Cleaning

Missing Value Handling

Checked missing values using:

- ☐ isnull()
- ☐ isna()

Numerical Columns

- ☐ Missing values were filled using median because median, mean is affected by outliers.

```
rainfall_df["actual"].fillna(rainfall_df["actual"].median(),inplace=True)
rainfall_df["rfs"].fillna(rainfall_df["rfs"].median(),inplace=True)
rainfall_df["normal"].fillna(rainfall_df["normal"].mean(),inplace=True)
rainfall_df["deviation"].fillna(rainfall_df["deviation"].mean(),inplace=True)
```

Duplicate Handling

- ☐ Checked duplicate rows using: duplicated()
- ☐ Removed duplicate records to avoid repeated analysis.

Datatype Conversion

The date column was converted into datetime format for time-series analysis

```
rainfall_df['date'] = pd.to_datetime(rainfall_df['date'])
```

A new month column was extracted from the date column.

Creating Rainfall Categories

Rainfall observations were categorized into different rainfall intensity levels

```
rainfall_df['Rainfall_Category'] = 'Low'

rainfall_df.loc[rainfall_df['actual'] > 50,
               'Rainfall_Category'] = 'Moderate'

rainfall_df.loc[rainfall_df['actual'] > 100,
               'Rainfall_Category'] = 'Heavy'
```

Outlier Analysis

Outlier Detection Methods

- ☐ Boxplot visualization
- ☐ Skewness analysis
- ☐ Kurtosis analysis Columns with Outliers
- ☐ All numerical columns showed significant outliers.

Why Outliers Existed

- some regions receive unusually heavy rainfall
- seasonal monsoon variations occur
- climatic changes create sudden rainfall spikes

Observation

- Actual - Highly right-skewed
- Rfs - Highly right-skewed
- Normal - Moderately skewed
- Deviation - Extremely high outliers

Purpose of Log Transformation

Log transformation helped:

- reduce skewness
- compress large values
- improve data distribution
- preserve important rainfall information

After transformation:

- distributions became more balanced
- histograms improved significantly

IQR Capping - Deviation

The deviation column contained:

- extremely high outliers
- both positive and negative values
- severe skewness

Log transformation was not suitable because logarithmic functions cannot properly handle negative values, Therefore, IQR capping was applied.

After capping:

- skewness reduced from approximately 161 to 1.369
- boxplot distribution improved significantly

Why Outliers Still Existed

- rainfall datasets naturally contain extreme weather events
- removing all outliers may remove genuine rainfall observations
- aggressive outlier removal can distort real-world patterns

Final Insights

- Rainfall patterns varied significantly across months and states.
- Most rainfall variables were highly right-skewed with extreme outliers.
- Log transformation reduced skewness and improved data distribution.
- Deviation contained the highest outliers and was treated using IQR capping.
- Positive correlation existed between actual, rfs, and normal rainfall.
- Moderate rainfall events occurred more frequently than heavy rainfall events.

Recommendations

- Use log transformation for highly right-skewed rainfall variables.
- Apply capping instead of removing genuine rainfall outliers.
- Monitor extreme rainfall deviations for climate analysis.
- Use monthly rainfall trend analysis for seasonal forecasting.
- Further predictive analysis can be performed using machine learning techniques.

Conclusion

The rainfall dataset was successfully cleaned, preprocessed, and analyzed using exploratory data analysis techniques. Missing values were handled appropriately, and skewed variables were treated using log transformation and IQR capping methods. Various visualizations helped identify rainfall trends, correlations, seasonal behavior, and outlier patterns. After preprocessing, the dataset became more suitable for analysis, interpretation, and future predictive modeling.

Log Transformation

- ☐ To reduce extreme positive skewness in engagement_rate.
- ☐ To compress very large engagement values.
- ☐ To improve visualization and statistical interpretation.
- ☐ To make the data distribution more stable and manageable.

Impact of Log Transformation

- ☐ Reduced skewness significantly.
- ☐ Improved distribution balance.
- ☐ Reduced the influence of extreme values.
- ☐ Improved readability of plots and statistical analysis.

Why Outliers Still Existed

- ☐ Some viral engagement values remained even after transformation because they represented genuine audience behavior.